University of Notre Dame

Department of Computer Science and Engineering

CSE 60745

Graph Learning: Techn & Apps

Assignment Date:	
Sumission Date:	
Student Name:	

CSE 60745 (Fall 2026) – HW3-Part(B)

* This assignment contains 70 points. You are encouraged to finish independently. Any submitted work that it copied from any source or too similar to be an independent write-up will not be given credit. Please submit this homework (as a PDF or Word file including the cover page) to ND Canvas by **11:59pm on Monday, 10/26/2026**.

In this assignment, you will practice the implementation of basic graph transformers. Please use Google Colab for experiments and **submit only your implemented code file**. Useful papers for this assignment are listed as follows:

[1] Chen, Dexiong, Leslie O'Bray, and Karsten Borgwardt. "Structure-aware transformer for graph representation learning." In ICML 2022.
[2] Ying, Chengxuan, Tianle Cai, Shengjie Luo, Shuxin Zheng, Guolin Ke, Di He, Yanming Shen, and Tie-Yan Liu. "Do transformers really perform badly for graph representation?." In NeurIPS 2021.
[3] Shirzad, Hamed, Ameya Velingker, Balaji Venkatachalam, Danica J. Sutherland, and Ali Kemal Sinop. "Exphormer: Sparse transformers for graphs." In ICML 2023.
[4] Rampášek, Ladislav, Michael Galkin, Vijay Prakash Dwivedi, Anh Tuan Luu, Guy Wolf, and Dominique Beaini. "Recipe for a general, powerful, scalable graph transformer." In NeurIPS 2022.
[5] Wu, Qitian, Wentao Zhao, Chenxiao Yang, Hengrui Zhang, Fan Nie, Haitian Jiang, Yatao Bian, and Junchi Yan. "Sgformer: Simplifying and empowering transformers for large-graph representations." In NeurIPS 2023.

Q1: Structural Encoding in Graph Transformer (20 pts)

Complete LapPETransformer() and SPDAttention() to implement graph Laplacian absolute structural encoding and shortest path distance-based relative structural encoding in graph transformer. You may refer to the demo in Section 4 for implementation ideas.

Q2: Efficient Graph Transformer (20 pts)

Complete SparseSPDAttention() to implement attention mask for sparse graph attention. You only need to specify the node pairs used for attention in your implementation. It is encouraged to test different masking strategies.

Q3: Hybrid Graph Transformer (30 pts)

Complete HybridGTLayer() to implement a hybrid graph transformer composed of both MPNN-based local attention and transformer-based global attention. You can implement GraphGPS or design your own hybrid structures.